Integrating Sound Scattering Measurements in the Design of Complex Architectural Surfaces

Informing a parametric design strategy with acoustic measurements from rapid prototype scale models

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Abstract. *Digital tools present the opportunity for incorporating performance analysis into the architectural design process. Acoustic performance is an important criterion for architectural design. There is much known about sound absorption but little about sound scattering, even though scattering is recognized to be one of the most important factors in predicting the acoustic performance of architectural spaces. This paper proposes a workflow for the design of complex architectural surfaces and the prediction of their sound scattering properties. This workflow includes the development of computational design tools, geometry generation, fabrication of test surfaces, measurement of acoustic performance, the incorporation of this data into the generative tool. The Hexagon Wall is included and discussed as an illustrative design study.*
Keywords. *Architectural Acoustics, Parametric Design, Rapid Prototyping.*

Introduction

Architects are interested in complex forms again. The emergence of new software and new fabrication techniques has made both the design and realization of these forms possible. There is a rise in the use of computation in architectural design, and with this, the ability of architects to include analysis in the digital tools they are creating. With better and more available engineering performance software available, these analysis packages are being adopted by architects in their design processes. (Kolarevic and Malkawi, 2005) Architects are able to design and build complex forms and to include performance analysis as part of the design process, see Figure 1. (Peters, 2007) Parallel to the development of new digital design tools has been the development of new digital fabrication techniques, both in terms of the construction of actual buildings and for the development of architectural design models. (Peters and DeKestelier, 2008)

An important aspect of a building's performance is its acoustic performance. Acoustic performance is an inevitable criteria of every architectural space. Wallace Sabine first demonstrated the importance of acoustic absorption in the early part of the twentieth century, and the calculation of acoustic absorption and reverberation time has been the cornerstone of

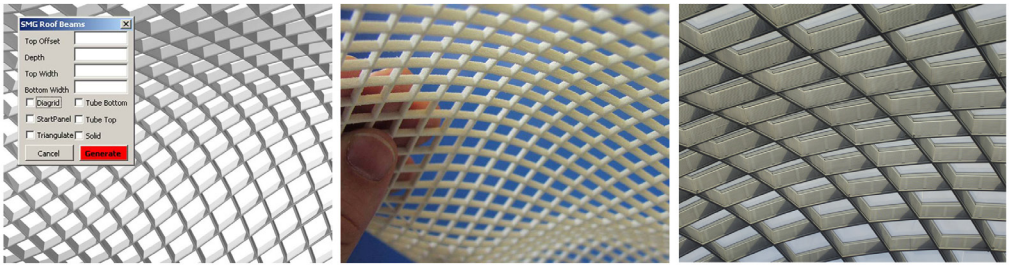


Figure 1
Smithsonian Courtyard by Foster + Partners: Digital Design Tool, Rapid Prototype Scale Model, and Acoustic Absorbing Roof Structure

architectural acoustics for the last 100 years. (Sabine, 1964) When a sound wave reaches a surface it can be absorbed, reflected, or scattered. Reflected sound can be either be redirected in a particular direction, or the sound can be scattered in many directions by a complex surface. While most architectural acoustics is concerned with sound absorption, sound diffusion is often important for obtaining an even distribution of sound. Sound diffusion is necessary for unevenly distributed absorptive materials to be effective, (Ushiyama et al., 2005) and acoustical defects such as flutter echo can develop in a room without adequate diffusing characteristics. (Long, 2006)

There is currently no architectural design software that includes sound as a design parameter (DeBodt, 2006). Therefore, current approaches by architects to designing for sound include the creation of custom digital design tools, see Figure 2. (Peters,

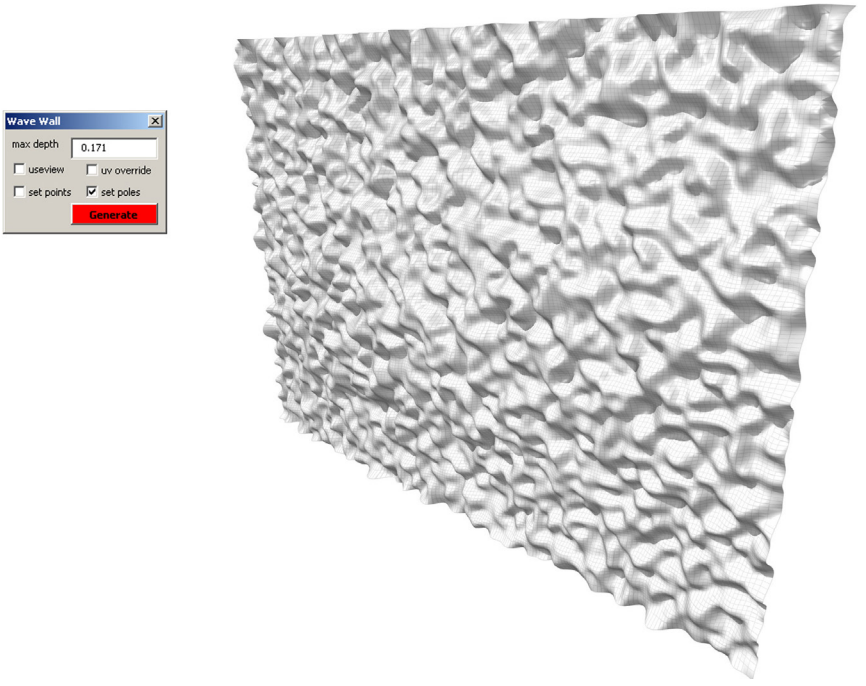


Figure 2
Wave wall acoustic scattering surface with parametric controls

2009, Koren, 2009, Bonwetsch et al., 2008) These approaches also allow for the integration of other types of performance analysis and the generation of fabrication data directly from the digital design tool.

In order to use sound as a design driver we must be able to measure and understand the acoustic consequences of our actions. By altering the geometry or material characteristics of the surfaces of a room, the acoustic properties are modified. Once the rules governing these acoustic alterations are understood, these rules can be encoded into a design system through parametric modeling or the use of computer programming. These tools then allow the architectural designer to design according to desired acoustic performance criteria. Design tools can be developed given known mathematical relationships of acoustic performance and geometric form, and these relationships can be encoded into an architectural design tool. Then the form can be explored from both acoustic and visual standpoints. (Peters, 2009)

One method to understand acoustic performance is to use room acoustic simulation and modeling software and incorporate its use into the architectural design workflow. The purpose of room acoustic modeling is to predict the acoustic performance of a room before it is built. One of the most challenging problems facing architectural designers and acousticians when designing indoor environments is the prediction of the overall sound quality before the design is committed to construction. Physical scale modeling has been applied as a prediction tool for over 75 years, but now computer simulation has offered a still more convincing alternative to scale models. (Rindel, 2002) It is commonly accepted that room acoustic simulation and modeling software based on geometrical acoustics must include both acoustic absorption coefficients as well as sound scattering coefficients in order to make reliable predictions of the acoustic condition. (Christensen and Rindel, 2005) The sound absorption properties of many materials are known and available through acoustic analysis software (Christensen, 2009), or can be found from material manufacturers. However, while the sound scattering properties of a few products are available, (RPG, 2009) for most architectural surfaces these material coefficients are not known. If new surface geometries are developed to be performative as a sound scattering device, their performance will be difficult to predict.

This paper proposes an architectural design workflow for the design of sound scattering surfaces through the use of parametric design tools, experimental measurement techniques, and the integration of the calculated material performance into the parametric design tool. Through the incorporation of acoustic data, complex architectural surfaces can become sound and space defining. Digital tools can be used to generate complex surfaces that have known scattering potential, see Figure 3.

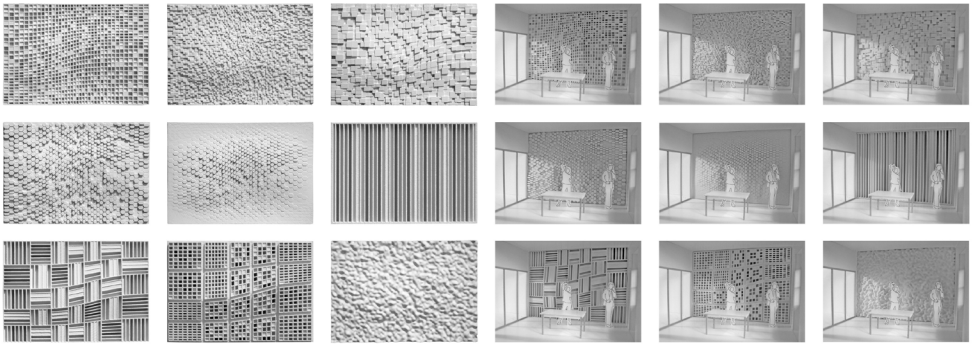


Figure 3
Rapid prototype models for acoustic measurement of scattering properties

While this process has been demonstrated for singular cases in previous work, (Luxemburg et al., 2002, Koren, 2009) where it is used as a validation tool to test a design concept, it has not been demonstrated as being part of a an iterative architectural design cycle. This research does not propose the design of a singular optimal surface and the testing of this singular case, instead this project proposes the development and testing of many, potentially radically different options, and integrating acoustic performance criteria as one of many architectural design criteria. This then allows acoustic performance to move from a validation procedure done late in the design process, to a conceptual design tool that will allow acoustics - and sound scattering performance - to become part of the conceptual architectural design strategy. The measured performance parameters are integrated into the generative computational model. This design workflow has application beyond architectural acoustics to other performance-driven design strategies and their relation to material testing. The integration of the test data allows the computational tools to "learn" from the material testing, and for future designs to then have increasingly predictable performance. This design strategy has been done previously, but for acoustic absorption properties, in the design of the Smithsonian Courtyard Enclosure. In this case, the material performance of the beam construction was studied in a reverberation chamber; the results of this analysis were then used to inform the digital model so it could be driven by the required acoustic performance. (Peters, 2007)

Theory of Sound Diffusion and Sound Scattering

The study of sound scattering and sound diffusion has benefited from an increased amount of exposure and research. (Cox and D'Antonio, 2009) While it is commonly recognized that sound scattering is important, there is still uncertainty as to how to design complex surfaces to achieve the desired acoustic performance characteristics. (Cox and D'Antonio, 2004) The terms scattering and diffuse reflections are often used interchangeably, however this paper will use established definitions as follows. (Cox et al., 2006) Scattering and the scattering coefficient is the measurement of the amount of sound scattered away from the specular reflection direction. Diffusion and the diffusion coefficient is a measure of the quality in terms of spatial uniformity of reflections produced by a surface. The scattering coefficient is what is used in computer simulation software, and is seen to be an adequate quantity for describing rough surfaces. It does not include any information about the directivity of the scattered energy, (Vorlander and Mommertz, 2000) as in room acoustical prediction methods, the directivity information can be described in terms of Lambert's cosine law. (Kuttruff, 2000)

Unfortunately, current acoustic analysis software cannot predict the diffusion or sound scattering coefficients. Sound scattering performance must be determined by either computational means or by the testing of physical models. (Cox 2009) Computational prediction of performance describes a version of nature through mathematical models, while the experimental testing of scale models or prototypes actually monitors the physical phenomena itself. Though computer simulation is becoming more widespread (Rindel, 2002) physical models have a role to play in acoustic design of complex rooms. (Gade, 2006) This project uses the method of the measurement of sound scattering from scale models. At present it has not been investigated in depth what scattering coefficient should be used for what materials. Odeon, an acoustic analysis software in widespread use, suggests five different material types and approximate values. (Christensen, 2009) It is left to the user to guess what and appropriate surface scattering coefficient is for their material. This research proposes a workflow by which the designer of surfaces can now make much more informed "guesses" as to the scattering performance.

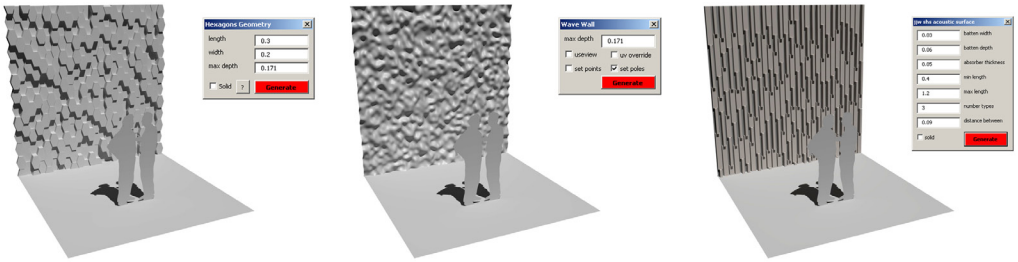


Figure 4
Design Tools and Wall Surfaces: Hexagon, Wave, and Batten Walls

Design of Sound Scattering Surfaces

Three different geometric configurations were studied. Three digital tools were developed that could generate the geometry of the different surfaces: the *Hexagon Wall*, the *Wave Surface*, and the *Random Battens* wall. The digital tools were developed in Microstation VBA. The digital tools take a bspline surface as an input and generate a diffusing structure on this surface. Of the three surfaces, the Hexagon Wall was studied in the greatest detail. The parameters that were studied were the well depth and well width of this surface type. The Hexagon Wall study will provide the focus for the rest of the paper.

The Hexagon Wall sound scattering strategy was loosely based on the Schroeder diffuser and the research of Cox and D'Antonio. (2009) A Schroeder diffuser consists of a series of wells having the same width but different depths. A mid-frequency sound wave incident on the diffuser surface will result in plane wave propagation within the diffuser wells. The waves are reflected and the pressure at an external point to the diffuser is determined by the interference pattern of the waves radiated from wells. The choice of the well depths determines the angular distribution of the reflected pressures. This is due to the equal magnitude (no energy loss) but different phase characteristics (due to different well depths) introduced by the diffuser. It has been found that by using different mathematical sequences, such as the quadratic residue sequence, a diffuser can exhibit quite uniform scattering. However, there are problems with the periodicity of the wells producing periodic diffusing effects. A random sequence of wells is less likely to exhibit this defect, but the scattering performance is much more difficult to predict using simple mathematics. The depths of the wells in a Schroeder diffuser vary between zero and a maximum depth of one-half of the wavelength of the design frequency. Along with the effect of the well depth on the scattering produced by the surfaces also the well width is varied. A larger well width reduces the number of wells and edges on the sample thus affecting the diffuser characteristics. (Cox and D'Antonio, 2009)

Measuring the Sound Scattering Performance

The measurement of the sound scattering coefficient followed the guidelines as established by the ISO 17497-1 standard, Measurement of the random-incidence coefficient in a reverberation room. (ISO, 2004) This standard defines the procedure by which a material sample, either full scale or partial scale, is placed on a turntable and impulse responses are obtained for different source and receiver positions in a reverberation room. The volume of the reverberation room must be at least 200 m3, or 0.20 m3 at 1:10. The volume of the reverberation room used was 0.45 m3. A uniform distribution of the natural frequencies is desired, and therefore no two dimensions in the reverberation room should have a ratio of small whole numbers. As seen in Figure 5 the reverberation room has an irregular shape and has no parallel walls. This avoids flutter echoes and increases the diffusivity of the room. The turntable used for rotating the model

has a circular base plate is symmetric with respect to the axis of rotation and is a rigid metal structure. The shortest distance between the base plate and a given room wall is 0.158 m which is larger than the required minimum of 0.10m. The equipment used for the measurements was a laptop containing the DIRAC Software, an E-MU 0202 USB sound device sampling up to 192 kHz, a B&K 1/4" microphone, a B&K Nexus amplifier, a 10W Amplifier, a custom 1:10 omni-directional loudspeaker, a B&K turntable 9640, and humidifier.

Calculation of the random-incidence scattering



coefficient is based on measurement of absorption coefficients. This necessitates the measurement of the reverberation time under four conditions: with and without the test surface present, and for rotating and non-rotating conditions. The impulse responses are measured indirectly using an exponential sine sweep. The sweep length is 2.73 seconds and a total of 72 coherent averages were conducted for each measurement. The DIRAC software's algorithm calculates the decay curve from the measured impulse responses. The reverberation time (T15) is then determined in the range between 5 dB and 20 dB. The impulse responses are measured for two sound source positions and three microphone positions giving a total of six spatial averages.

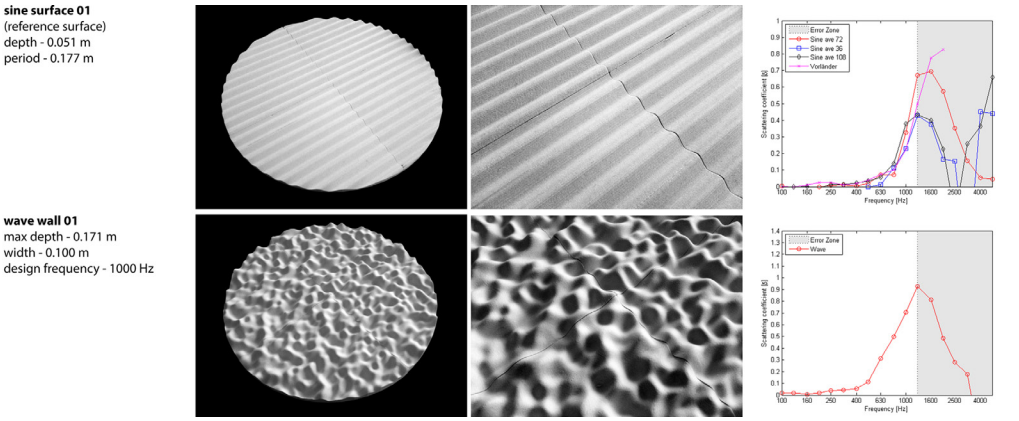
Results and Discussion

The different design options were generated using the parametric tools previously described specifically to study the effects of varying the parameters well-depth and well-width. 1:10 models were fabricated using a ZCorp 3d printer. According to the standard (ISO, 2004) the area of the test sample should be as large as possible and circular with a minimum diameter of 0.30 m at scale 1:10. Due to the bed size of the machine, the models were produced in four pieces, which come together to form a single piece with a diameter of 0.354 m. The structural depth of the samples should be smaller than or equal to 1/16 of the diameter. The ISO method is intended for the measurement of surface roughness. The absorption coefficient of the test sample should not exceed 0.5, however, it should still be present if absorption is part of the surface construction. A flat sample piece was measured to determine the absorption coefficients of the rapid prototype material (ZP150 powder). It was found not to have significant absorptive characteristics. The second measurement done was of a sine surface. This surface is the same dimensions that has been used in previous published experiments by Vorlander, et al. (2004) and was used as a reference to establish the validity of the results. The values measured for the scattering coefficient as seen in Figure 6 are in agreement with the reference surface up to 1250 Hz, after which there is a divergence of results. The results of all of the other measurements also show inaccuracies after 1250 Hz. We are therefore uncertain of measurements above 1250 Hz.

Six different options of the Hexagon Wall surface were modeled and tested. This corresponded to four different well-depth conditions, and three different well-width conditions. Figure 7 shows two views of the 1:10 model and the measured scattering coefficient expressed over a frequency range of 100 Hz to 4000 Hz. All surfaces conform to the ISO standard dimensions except for hexagon04, which exceeds the recommended structural depth. All of the surfaces have a high degree of detail and very

Figure 5
1:10 Reverberation Room with Turntable, Omnidirectional Loudspeaker, Microphone, and Computer

Figure 6
Reference Surface and Wave Surface- 1:10 Rapid Prototype Scale Models and Calculated Scattering Coefficients



complex geometry which affects the scattering properties. The frequency characteristics of the measured results for the different surfaces demonstrate that an increase in well-depth increased the scattering coefficient at lower frequencies as predicted. The amount of scattering is quite high for all options except for hexagon01, the option with the least well-depth. However, as the design frequency for this option is 4000 Hz, which is above the accurate range of our experiments as explained previously, the measurement equipment may simply have not captured the scattering activity at the high frequencies above 1250 Hz. The widest option, hexagon06, also had a decreased value for the scattering coefficient. This corresponds to a decrease in the amount of detail of the complex surface. It can be seen that the well-depth and amount of detail impact the scattering coefficient for these design options.

One of the most challenging aspects of this workflow is the measurement process. It is the opinion of the authors that better guidance on this process should be given if it is to be used as part of the design workflow. When applying the numerical method for compensating for air absorption, the background noise is amplified dramatically at high frequencies (Choi and Jeong, 2008), this may have been a problem in our measurements. Issues with humidity and air absorption have been identified as a key issue in other experiments, so it is critical to carefully monitor and maintain constant humidity and air temperature. Another possible source of error could be the lack of diffusion in the reverberation room. The reverberation room also has a large volume (though within the standard) which makes the air absorption even more of an issue. For further experiments we would suggest diffusers be added to the reverberation room. As previously discussed, there was a problem with the measurements above 1250 Hz. If measurements above 1250 Hz were possible, smaller geometries, producing higher frequency scattering could be measured.

Diffraction at the edge of the sample can effect the measurement accuracy. This is why the ISO standard dictates a height of no more than 1/16 the diameter of the sample. Edge effects prompted by variations of the sample height along its edge can result in scattering coefficients greater than those which would otherwise be obtained if the sample were an infinite surface area. It has been found that the reliability of the measurement can be improved by inserting the test sample into the floor of the reverberation chamber, so that it is flush with the floor level, reduces the edge effect. (Lin et al., 2010) This may also allow the testing of geometries greater than 1/16 the diameter of the sample. This would allow larger geometries that would scatter sounds at

hexagon 01
max depth - 0.043 m
width - 0.092 m
length - 0.069 m
frequency - 4000 Hz

hexagon 02
max depth - 0.085 m
width - 0.092 m
length - 0.069 m
frequency - 2000 Hz

hexagon 03
max depth - 0.171 m
width - 0.092 m
length - 0.069 m
frequency - 1000 Hz

hexagon 04
max depth - 0.342 m
width - 0.092 m
length - 0.069 m
frequency - 500 Hz

hexagon 05
max depth - 0.171 m
width - 0.184 m
length - 0.139 m
frequency - 1000 Hz

hexagon 06
max depth - 0.171 m
width - 0.368 m
length - 0.278 m
frequency - 1000 Hz

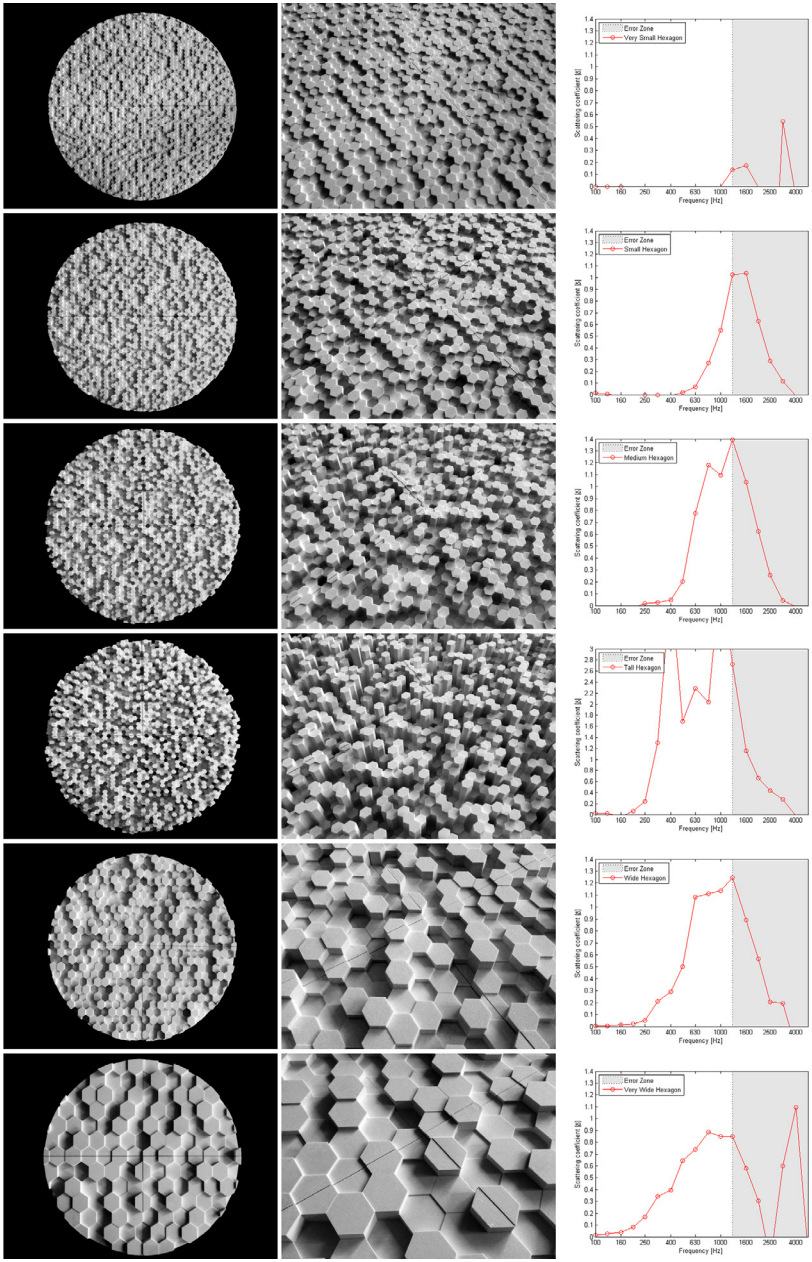
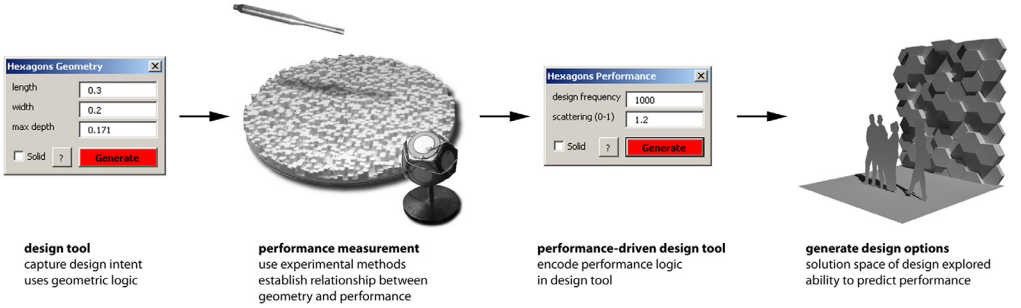


Figure 7
Hexagon Surface Structures varying by Depth and Width: 1:10 Rapid Prototype Scale Models and Calculated Scattering Coefficients

Figure 8
Design Process for Creating Performance-driven Design Tools



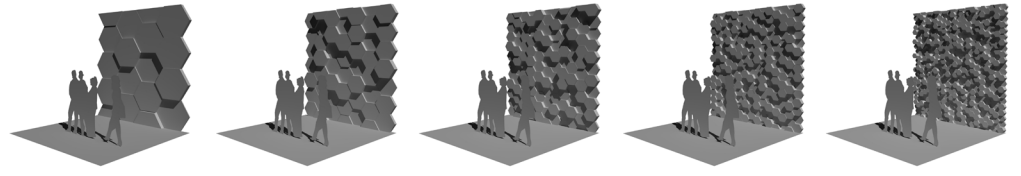
lower frequency structures to be tested. We would suggest further measurements are mounted flush with the floor to remove the edge effect.

Integration of Performance Results into the Design Tool

The analysis of the measured results demonstrates trends that relate well-depth to the frequency distribution of the scattering coefficient and the well-width to the magnitude of the scattering coefficient. These relationships were then encoded into a new version of the digital design tool, and further design options were produced. The design workflow is illustrated in Figure 8. This workflow allows the design of geometries with predictable sound scattering properties. Thus the design tool can control the amount of scattering, and frequency of scattering. Because of the previously discussed limitations of the frequency range of the measurement apparatus and the geometric relationships established, an iterative test in was not done. As previously discussed, one of the main reasons for the development of the scattering coefficient is for its use in computer simulation programs; therefore, it is important that the results of the measurements and of the design tool are also compatible with acoustic simulation software. While the ISO standard procedure generates results over one-third octave band frequencies, Odeon acoustic software only uses a single number as input. Therefore the ability to accurately capture the particular frequency characteristics of the scattering properties of the surface may not be entirely captured by the single number value. If computer simulation software is to agree with the ISO standard, it should incorporate frequency dependent results obtained by the standard, or the ISO standard should be changed, to provide guidance on how to achieve the correct single value scattering coefficient.

With this design workflow, decisions about geometry can be made not only using visual criteria, but also using sonic criteria. This design tool gives

Figure 9
Hexagon Wall Surfaces Generated by Performance Criteria



instantaneous feedback to the designer in terms of what performance can be expected from the designed surface. However, it does assume that the designer has knowledge about what this scattering coefficient means and how it affects the acoustic quality of a space. The design tool generates design options and allows for a fast and efficient exploration of the solution space. Several design options shown in Figure 9 were developed to illustrate different scattering properties at a common sound frequency.

Conclusions

In this research project, computational tools have been developed to incorporate the acoustic parameter of sound scattering, to output data sufficient for rapid prototyping, and to output data suitable for import into acoustic analysis package. The ability to integrate sound scattering performance data into the parametric tool, and the testing of the newly generated surfaces, demonstrates a feedback system of informed tool-building and testing that can produce surfaces with more predictable sound scattering performance, and therefore allow architects to better design acoustic environments. The incorporation of performance data into the design tool is a great benefit to architects, however the difficulties encountered with the measurement of the scattering coefficient and its subsequent integration into simulation software indicates that more research needs to be done in this area.

This proposed design workflow is not intended to provide a fine-grain analysis, but to provide designers with computational tools that allow them to establish a closer relationship to the phenomena of sound. These tools are not intended to replace acoustic consultants. The techniques here are useful to incorporate acoustic logic into design tools - a way for design tools to "learn" from physical measurements. Instead of building on established acoustic mathematical relationships, this workflow allows for the understanding of the relationship between geometry and performance and the encoding the material behavior of the sound signal into the digital tool. The authors would like to thank Dr Cheol-ho Jeong and Dr Jonas Brunskog, Claus Moller Petersen, RealDania, and JJW Architecture for their support of this research.

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